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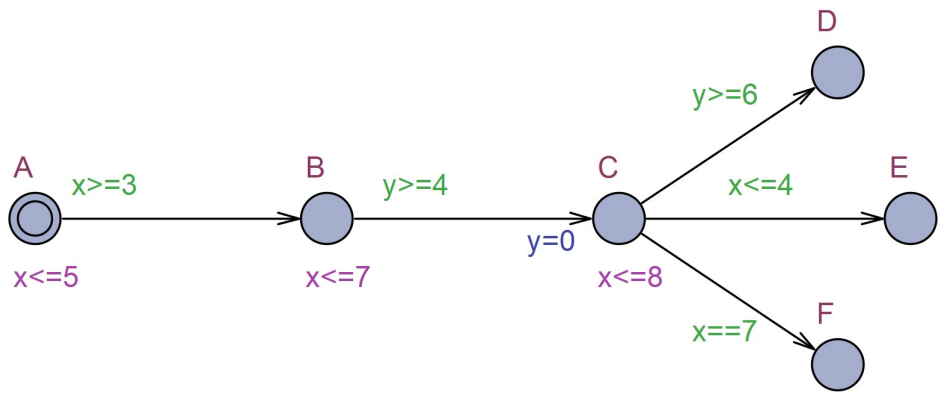
Models and Tools for Cyber-Physical Systems
Exercise sheet 7

Exercise 1: Symbolic Exploration of Timed Automata

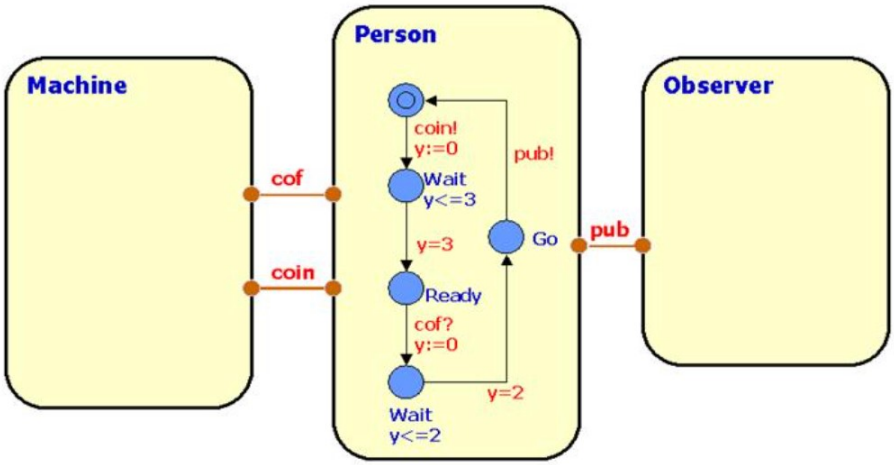
Consider the timed automaton $\mathcal{A}$ below with two clocks `x` and `y`. Perform a symbolic zone-base reachability analysis of $\mathcal{A}$ as in Lecture 6 (slides 37-44). Which of the locations `D`, `E` and `F` are reachable? Describe, using difference constraints, the reachable zones upon entry and after delay for the locations `A`, `B` and `C`.



You can find a Uppaal model of $\mathcal{A}$ in SimpleTA.xml in Moodle.

Exercise 2: Coffee Machine

In this exercise you are asked to design the control of a `Machine` (the control program) which will serve a coffee craving `Person` (the environment). As you can see below the person repeatedly (tries to) insert a `coin`, (tries to) extract `cof`fee after which (s)he will make a `pub`lication. Between each action the `Person` requires a suitable time-delay before being ready to participate in the next one.



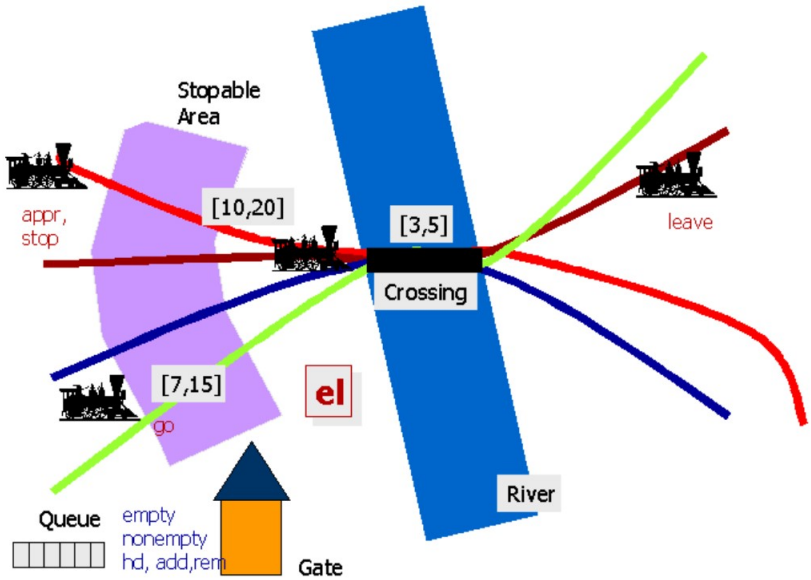
The `Machine` takes some time for brewing the coffee and will time-out if coffee has not been taken before a certain upper time-limit.

As a requirement we want the overall behaviour to ensure that the indicated `Observer` experiences a constant flow of publications from the system. In particular we want the `Observer` to complain if at any time more than 8 time-units elapses between two consecutive publications. Model the `Machine` and `Observer` in UPPAAL and analyse the behaviour of the system. Try to determine the maximum brewing time allowed by the `Machine` in order that the above requirement is met.

An initial sketch of a UPPAAL can be found in CoffeeMachine.xml in Moodle.

Exercise 3: Train-Gate Error Correction

Below is the overview of the `TrainGate` model. An erroneous UPPAAL model of this case may be found in train-gate-err.xml on Moodle and in train-gate-err.q you will find the properties.



Please do the following.

(a) Use UPPAAL (simulation, verification, or whatever you want) to pin-point and explain the error(s).

(b) Having corrected the errors, what is the minimum time one can guarantee between a train requesting access to the crossing and actually getting there?

(c) Having corrected the errors, what is the minimum time one can guarantee between a train being stopped and getting the permission to start again (i.e., the waiting time)?